William Rowan Hamilton — *Elements of Quaternions* (1866)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 14 October 2026.

Disclosure: prepared with AI assistance — the subject reviewers are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To William Rowan Hamilton (1805–1865), whose last book outgrew both his plan and his strength. He had allowed two years and about 400 pages for the Elements of Quaternions; he spent seven years, the book grew to roughly twice the intended length, and its final chapter was still incomplete when he died. It reached its readers in 1866 through the editorship of his son, William Edwin Hamilton, and was later enlarged by Charles Jasper Joly. This edition is dedicated to Hamilton, with the incomplete chapter marked as incomplete and each later hand credited for the work that is his.

The author's own terms

Quaternions are Hamilton's system of numbers with four components, developed for working with rotations and quantities in space.

The story

William Rowan Hamilton (1805–1865) spent seven years on the book he had planned to finish in two. It grew to roughly double the length he had intended, about 400 pages becoming something nearer twice that, and the final chapter was still incomplete when he died. The book was published in 1866, after his

death, with a preface by his son, William Edwin Hamilton, who edited it.

That is a human story before it is a bibliographical one. A man's last book outgrows his plan, outlasts his strength, and is carried into print by his son. The *Elements of Quaternions* — quaternions being Hamilton's system of numbers with four components, developed for working with rotations and quantities in space — reached its readers in that completed-by-another-hand form.

This dossier, however, is not about translation, and it must not be mistaken for one of the translation cases in this collection. Hamilton wrote in English. There is no translation gap, and none should ever be claimed.

The opening here is editorial. What English readers lack is not Hamilton's book in their language. What they lack is a modern edition of it that they can read, search and use easily today.

The work and what survives

The first edition of the *Elements of Quaternions* was published in 1866, edited by the author's son. A copy of that first edition is held, for example, at the Huntington, under call number 707525.

A second edition followed, enlarged by Charles Jasper Joly, in two volumes, published in 1899–1901. That edition was reprinted by Chelsea in 1969. The Internet Archive item `elementsquatern01hamigoog` is the Chelsea and Joly edition, Google-digitised from Harvard, with full OCR — optical character recognition, the machine-readable text taken from the page images.

The texts of 1866 and of 1899–1901 are in the public domain.

Against that survival, the present position is narrow. No modern critical or reset English edition found after research. What the research found instead was only the 1969 Chelsea photographic reprint of the 1899–1901 Joly edition, and print-on-demand facsimiles — photographic copies of old pages, reproduced as they stand.

A photographic reprint preserves a book, but it does not reset it. Its type stays as it was set more than a century ago, its pages cannot be searched as clean text can, and the different hands in the book are not easily distinguished by a modern reader.

Its English fortunes

Hamilton's English fortune is the opposite of the other cases in this part of the collection. His words have always been English words. Their difficulty is not language, but form: an unfinished final chapter, a son's editorship, an enlarged second edition by another editor, and, since then, reprints that reproduce those layers without clarifying them.

What completing it would take

What is needed is a clean, searchable, modern-typeset *Elements*, prepared from the public-domain scans. In such an edition, the incomplete final chapter would be clearly marked as incomplete, and the son's and Joly's editorial layers would be clearly marked and credited.

That crediting matters. William Edwin Hamilton and Charles Jasper Joly each did real editorial work on this book, at different times and in different ways. A new edition should let a reader see what Hamilton wrote, what his son did to bring the unfinished book to publication in 1866, and what Joly added in enlarging it in 1899–1901. Marked honestly, those layers do not weaken the book. They tell the truth about how a great, unfinished work reached the world.

Sources

- http://physics.gmu.edu/~isatija/phys303/Hamilton.html (seven years; incomplete final chapter; son's editorship)
- https://arxiv.org/pdf/math-ph/0201058v3 (Joly edition history)
- https://archive.org/details/elementsquatern01hamigoog
- https://www.huntington.org/collections/lib-707525